



UNIVERSITI TEKNOLOGI MALAYSIA

TASK 4

NETWORK COMMUNICATION

NO	NAME	MATRIC NO
1	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250
2	AHMAD ADIB ZIKRI BIN A.MAZLAM	A23CS0205
3	SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
4	YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201

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4.1 Identify the work areas on floor plan

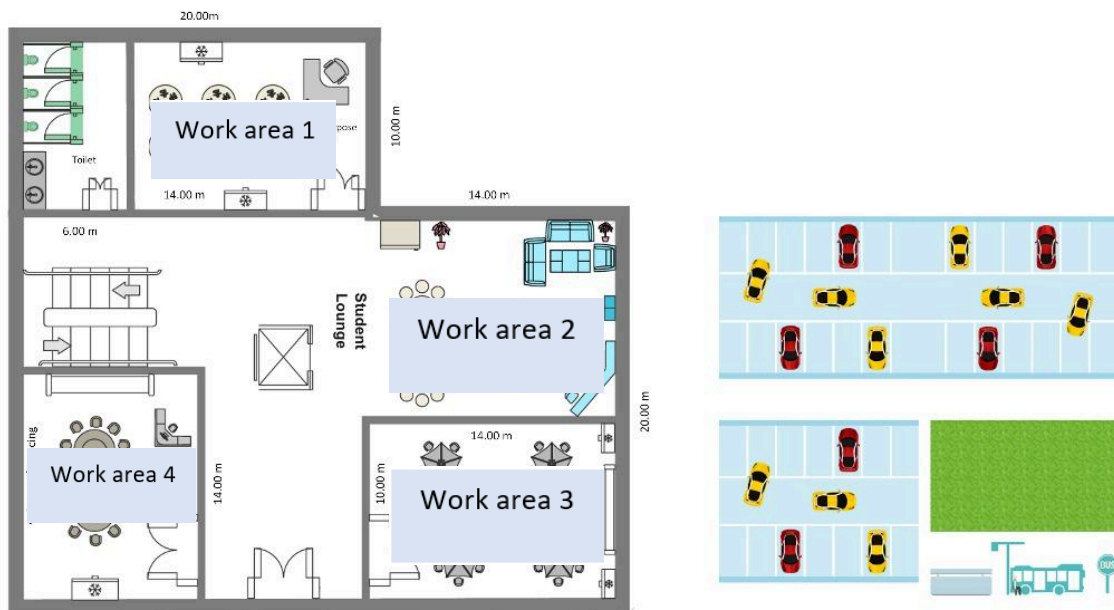


Figure 4.1.1 work area at first floor

Figure 4.1.1 depicts Work Areas 1, which are general labs 1 with dimensions of 14 meters by 10 meters. Students are supposed to use these general laboratories in class. There are 30 PC workstations in the lab. As a result, three 30-port switches are needed in the general labs to link every workstation.

Next is the work area 2 which is student lounge measures 14 meters by 10 meters and is designated as a collaborative and relaxation space for students. While the primary purpose is informal group discussions and socializing, networking is supported through a wireless access point, ensuring seamless internet connectivity for students using their personal devices. This area is equipped with couches, round tables, and bar-style seating for individual or group work, making it essential to have strong Wi-Fi coverage to accommodate the number of simultaneous connections typically seen in such spaces.

After that the work area 3 which is a hybrid classroom spans 14 meters by 10 meters and supports both in-person and remote learning environments. It is equipped with network connectivity for AV equipment and video conferencing systems, ensuring smooth communication between remote and in-class students. To handle the needs of approximately 25 workstations or devices, the classroom would require two 25-port switches for wired connections to interactive displays, projectors, and possibly some student devices. Additionally, a dedicated access point ensures consistent wireless performance for those accessing digital learning resources.

Then is the work area 4 which is the video conferencing room is 14 meters by 8 meters, designed specifically for remote meetings and communication. It is equipped with a robust network setup, including video conferencing hardware, to facilitate virtual meetings. For optimal performance, the room likely utilizes a 24-port switch to ensure high-speed connections for audio-visual devices, a central conferencing table, and any participant laptops or workstations. Adequate cabling and strong Wi-Fi support ensure that communication remains smooth, avoiding any latency or interruptions during virtual sessions.

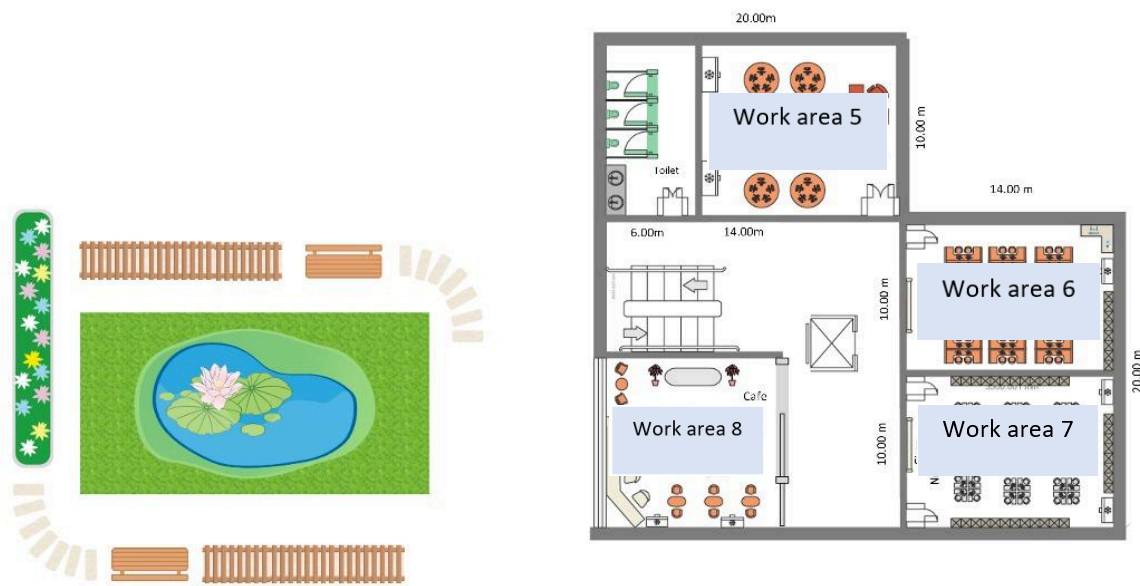


Figure 4.1.2 work area at second floor

Figure 4.1.2 depicts Work Areas 5 , which are General Purpose Lab 2 is 14 meters by 10 meters and designed for student use during class sessions. It houses approximately 30 workstations, making networking a key component for connectivity. To support these devices, the lab would require three 24-port switches for stable wired connections, ensuring every workstation has access to the network. Wireless access points may also be included for students who need to connect additional devices or work in a flexible environment.

Next, the work area 6, the Embedded Lab is 14 meters by 10 meters and focuses on providing students with specialized equipment and workstations for embedded systems projects. With around 20 workstations visible, at least two 24-port switches are necessary to handle the wired network requirements. In addition to the wired connections, wireless connectivity may be available for programming and debugging embedded devices, making a reliable network essential for both the students and instructors.

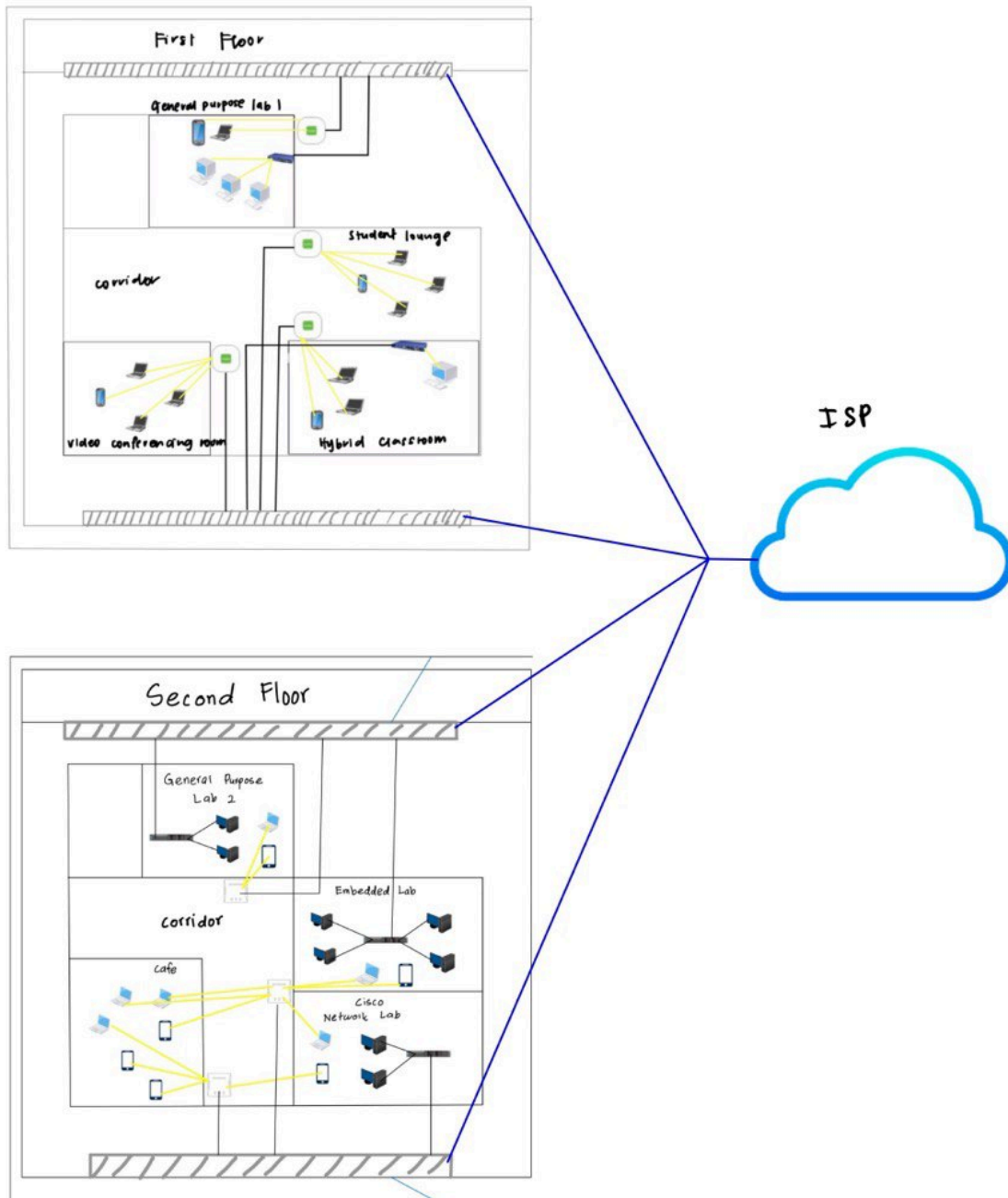
For the work area 7, The Cisco Network Lab measures 14 meters by 10 meters and is dedicated to networking courses and hands-on practice with Cisco equipment. This room contains numerous PCs and networking devices, each of which requires a network connection for simulations and experiments. The lab would likely need four 24-port switches, as there are more than 30 stations, along with routers and other networking equipment for educational purposes. High-speed internet access and local network configurations are essential to provide students with practical networking experience.

Lastly, work area 8 is the café area, with dimensions of 10 meters by 10 meters, serves as a social space for students and staff. While its primary function is casual dining and socializing, the café is equipped with wireless access points to provide network connectivity. This allows students to use their personal devices, such as laptops and smartphones, while they take a break, ensuring they stay connected to the internet for study or leisure activities.

4.2 The distribution and connection of devices

4.2.1 Network Distribution Diagram

Network Distribution Diagram



4.2.2 Room Level Network Distribution Diagram

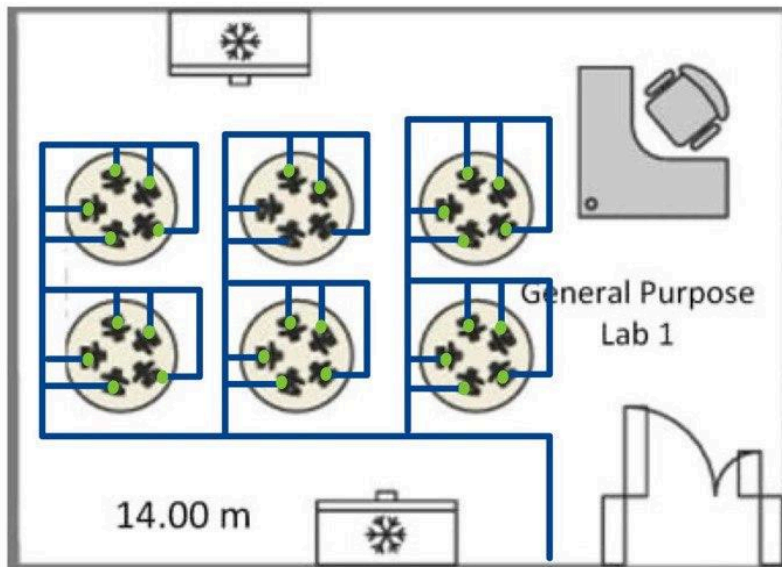


Figure 4.2.2.1 General Purpose Lab 1

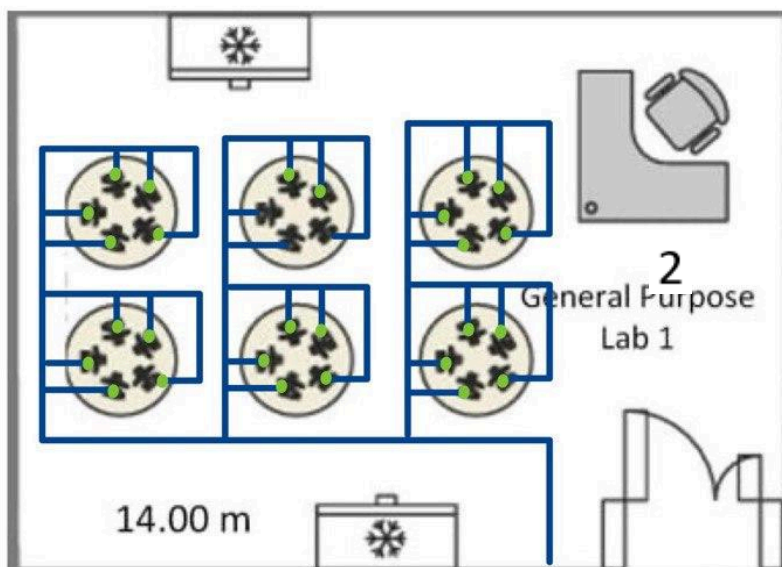
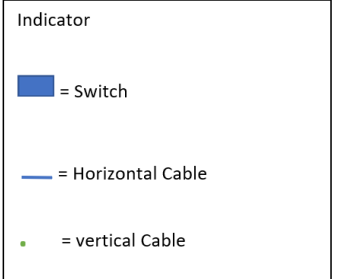


Figure 4.2.2.1 General Purpose Lab 1



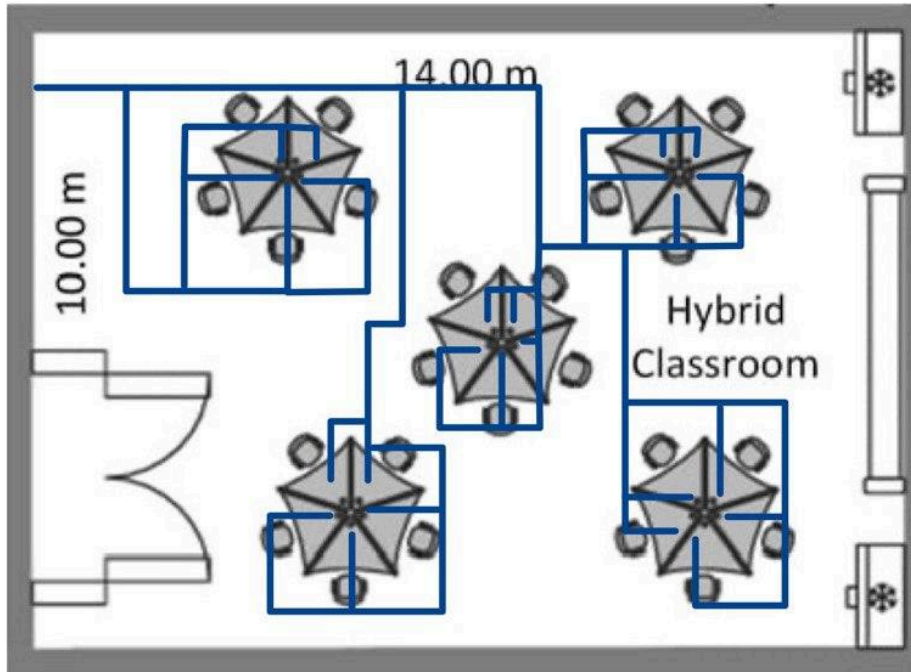


Figure 4.2.2.2 Hybrid Classroom

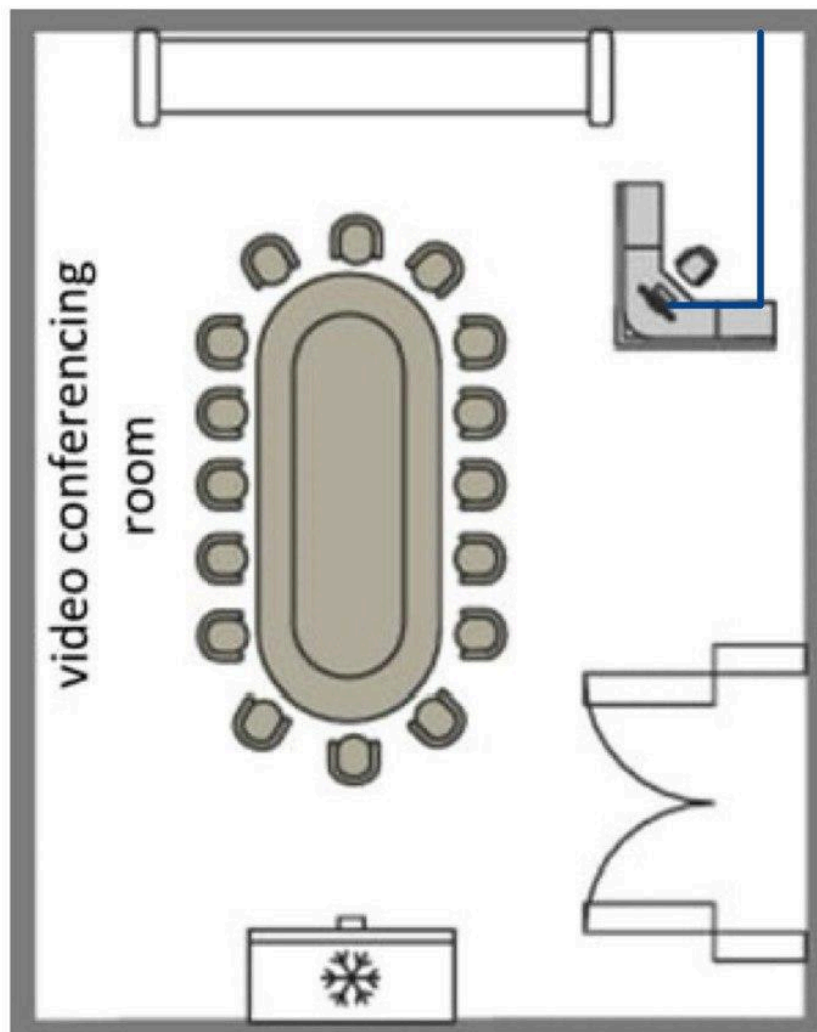
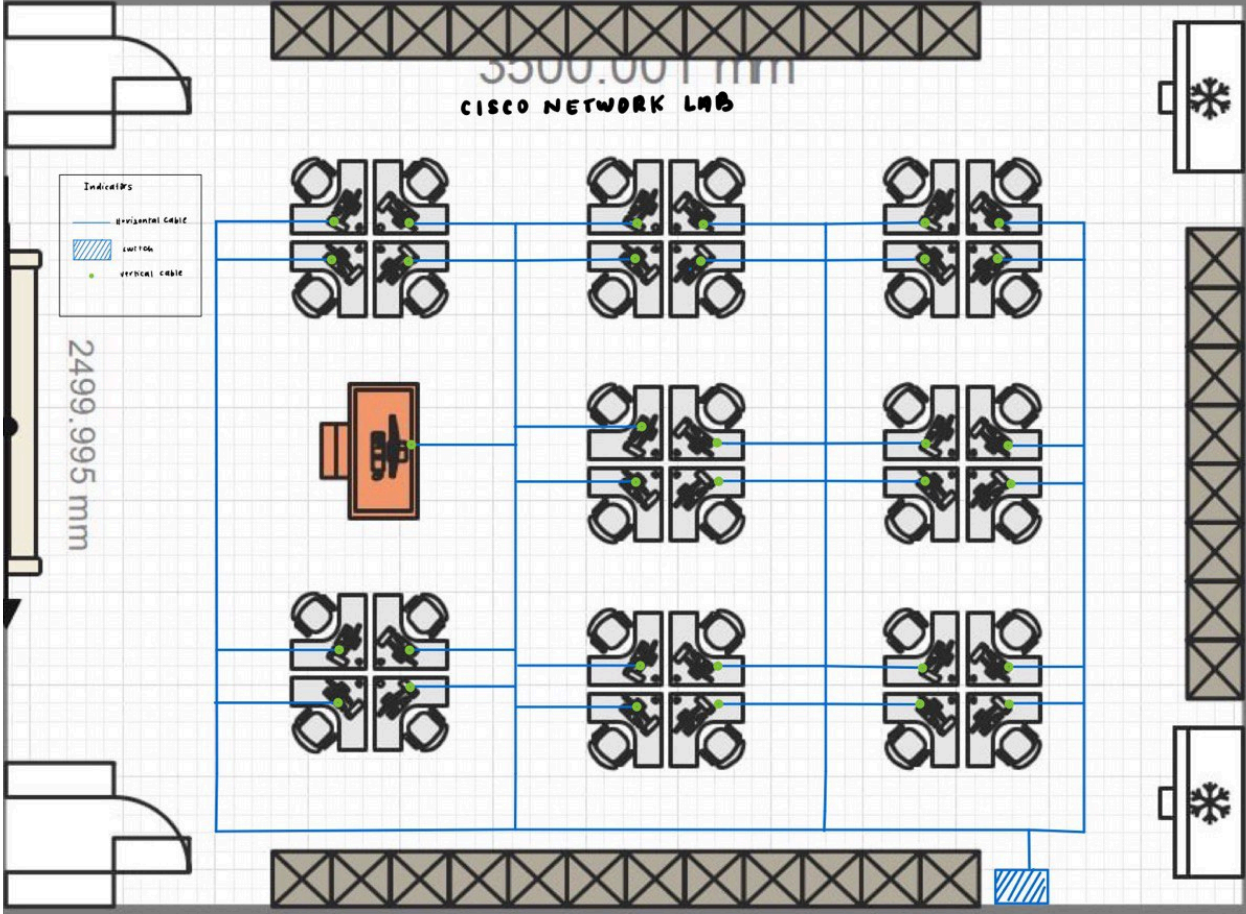
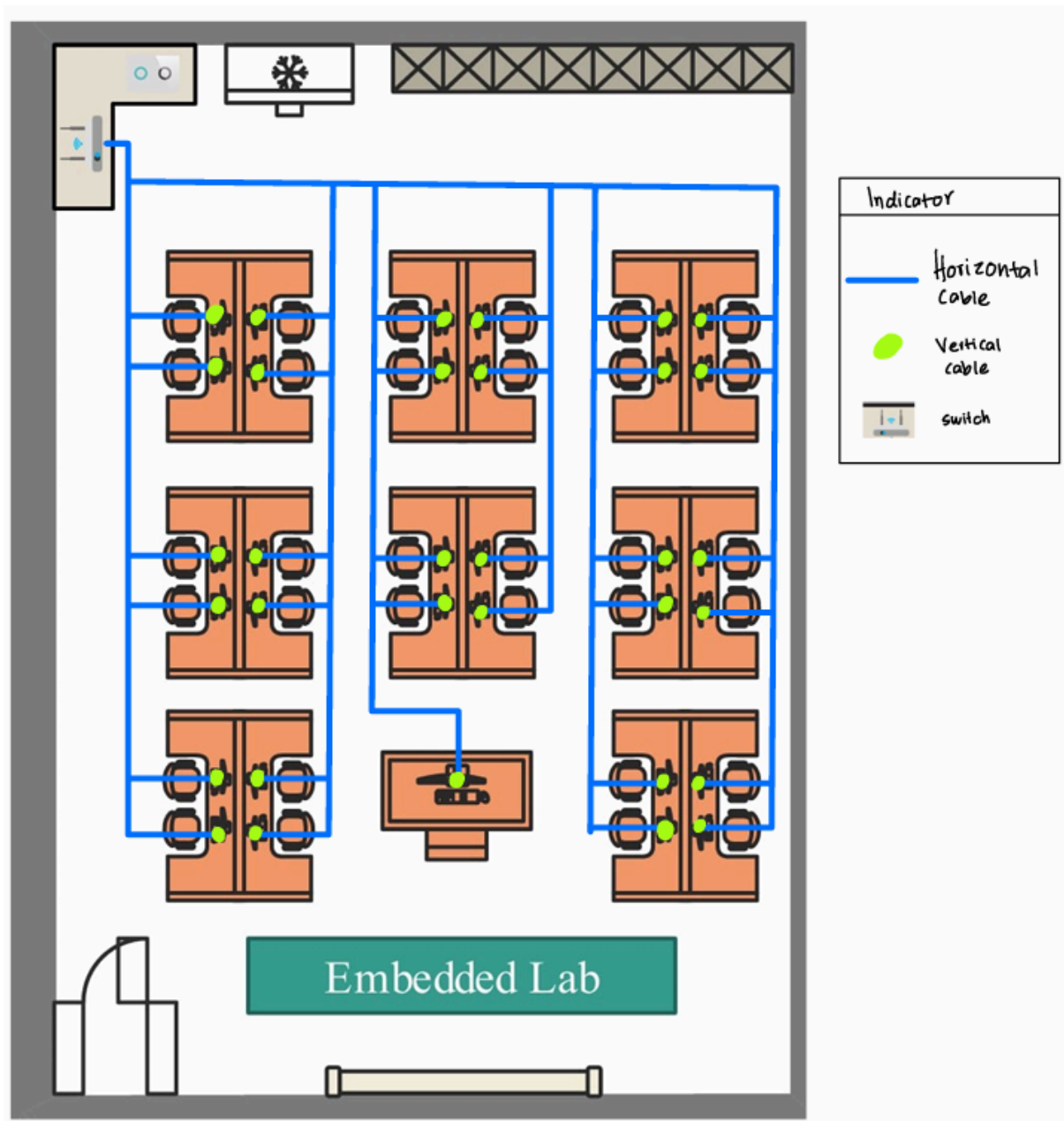


Figure 4.2.2.3 Video Conferencing room





4.2.3 Floor Diagram

4.3 Number of connection, patch cords and switch ports needed

Area	Number of switch ports needed
General Purpose Lab 1	30
General Purpose Lab 2	30
Video Conferencing Room	1
Hybrid Classroom	25
Embedded Lab	33
Cisco Network Lab	33

Table 4.3.1: Number of switch ports needed

Area	Number of patch cords needed
General Purpose Lab 1	30
General Purpose Lab 1	30
Video Conferencing Room	1
Hybrid Classroom	25
Embedded Lab	33
Cisco Network Lab	33

Table 4.3.2: Number of patch cords needed

Area	Number of RJ45 Connector needed
General Purpose Lab 1	60
General Purpose Lab 1	60
Video Conferencing Room	2
Hybrid Classroom	50
Embedded Lab	66
Cisco Network Lab	66

Table 4.3.3: Number of RJ45 Connector needed

4.4 Identify cable types and length

Area	Length		Total Length (m)	Total price of cable used (RM)
	Horizontal	Vertical		
First Floor				
Room				
General Purpose Lab 1	478.5	15	493.5	3311.39
Video Conferencing Room	5	3	8	53.68
Hybrid Classroom	383.5	7	390.5	2620.25
First Floor Room Total			892	5985.32
Second Floor				
Room				
General Purpose Lab 2	478.5	15	493.5	3311.39
Embedded Lab	195	40	235	1576.85
Cisco Network Lab	235.4	53.2	288.6	1936.51
Second Floor Room Total			1017.1	6824.75

In the previous section, we explained the strategy for distributing network connections and linking devices. To enhance connectivity, it's important to choose the appropriate cable types. After thorough group discussions, we have decided to use a mix of CAT6 twisted pair cables and fiber optic cables throughout the building.

The alternative approach selected for the network setup in the labs and lecture halls was with CAT6 cables to ensure flexibility and future-proofing. CAT6 cables are chosen not only for their capability to handle high-speed internet but also because they are cost-effective in the

long run, whereby in the future, any upgrading of the network will easily be supported without the need for immediate recabling. Their robust design makes them perfect for environments that require consistency in data transmission, as it reduces the degradation of performance. In addition, utilizing CAT6 for both data and PoE contributes to simplifying installation; network devices can be powered up and connected using the same line, which reduces hardware clutter and adds to the aesthetic value of the workspace. The UTP variant had been chosen because it is the cheapest option that would provide sufficient shielding against interference, ensuring smooth data flow even in highly active areas with potential electromagnetic disturbances. This approach ensures a lean and scalable network infrastructure that is able to grow as the demands of technology increase.

MEETING MINUTE 1

DATE/TIME		15th December 2024	
LOCATION		ONLINE - GMEET	
AGENDA		1) Discuss the question given for Task 4 2) Divide each room for network distribution	
Meeting MC		MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		8:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		8:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		8:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		8:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the materials given for Task 4	i) Syarifah Dania suggested all of the team members go through the questions and determine the requirements needed to complete Task 2. ii) All members read the materials given and identify the requirements	All members (15/12)
2	Divide each room for network distribution	i) Yasmin Batrisyia and Ahmad Adib Zikri suggest each member draw the network connection for each room.	Yasmin Batrisyia (15/12)

		ii) All members draw network connections for each room.	Muhammad Mukhritz Al Iman (15/12)
3	Next meeting	i) All members agreed to conduct another meeting on 18th December as everybody is currently busy with programs ii) Agenda of the next meeting was decided. iii) Syarifah Dania reminded all members to do some research about the project to ensure that everyone will have more understanding before next meeting	All members (15/12)
5	Meeting ended	After completing the dividing task between each member, Ahmad Adib ended the meeting.	Ahmad Adib (15/12)

MEETING MINUTE 2

DATE/TIME	8nd of November 2024	
LOCATION	Ar-Rayyan Cafe, KTDI	
AGENDA	1) Analyse question from each member 2) Answer the question 3) Make a feasibility 4) Finalise the report.	
Meeting MC	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE

MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		9:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		9:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		9:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		9:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Analyse the drawing of network distribution	i) Muhammad Mukhritz prepared the drawing of network distribution prepared by each member ii) All members discuss and see if the distribution is suitable for the room	Muhammad Mukhritz Al Iman (18/12)
3	Reporting	i) Task distribution for the report was done ii) Cover page, table of content and Introduction were done by Muhammad Mukhritz. iii) Meeting minutes was done by Syarifah Dania.	All members (18/12)
4	Meeting ended	Yasmin Batrisyia ended the meeting once the report was completed.	Yasmin Batrisyia (18/12)

